Monomial f(q) = a*qˆ8 Report

Polynomial

$$f(q) = aq^8$$

Naive

$$g_{\mathrm{naive}}(X) = ax^8$$

$$\mathbb{E}[g_{\mathrm{naive}}(X)] = aq^8 + 28aq^6\sigma^2 + 210aq^4\sigma^4 + 420aq^2\sigma^6 + 105a\sigma^8$$

$$\begin{aligned}\mathrm{Var}(g_{\mathrm{naive}}(X)) = {} & 64a^2q^{14}\sigma^2 + 4256a^2q^{12}\sigma^4 + 107520a^2q^{10}\sigma^6 \\ & + 1283520a^2q^8\sigma^8 + 7385280a^2q^6\sigma^{10} \\ & + 18698400a^2q^4\sigma^{12} + 16128000a^2q^2\sigma^{14} + 2016000a^2\sigma^{16}\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\mathrm{naive}}) = {} & 64a^2q^{14}\sigma^2 + 5040a^2q^{12}\sigma^4 + 119280a^2q^{10}\sigma^6 \\ & + 1351140a^2q^8\sigma^8 + 7567560a^2q^6\sigma^{10} \\ & + 18918900a^2q^4\sigma^{12} + 16216200a^2q^2\sigma^{14} + 2027025a^2\sigma^{16}\end{aligned}$$

$$\mathrm{Bias}(g_{\mathrm{naive}}) = 28aq^6\sigma^2 + 210aq^4\sigma^4 + 420aq^2\sigma^6 + 105a\sigma^8$$

Unbiased

$$g_{\mathrm{unb}}(X) = 105a\sigma^8 - 420a\sigma^6x^2 + 210a\sigma^4x^4 - 28a\sigma^2x^6 + ax^8$$

$$\mathbb{E}[g_{\mathrm{unb}}(X)] = aq^8$$

$$\begin{aligned}\mathrm{Var}(g_{\mathrm{unb}}(X)) = {} & 64a^2q^{14}\sigma^2 + 1568a^2q^{12}\sigma^4 + 18816a^2q^{10}\sigma^6 \\ & + 117600a^2q^8\sigma^8 + 376320a^2q^6\sigma^{10} \\ & + 564480a^2q^4\sigma^{12} + 322560a^2q^2\sigma^{14} + 40320a^2\sigma^{16}\end{aligned}$$

$$\begin{aligned}\mathrm{MSE}(g_{\mathrm{unb}}) = {} & 64a^2q^{14}\sigma^2 + 1568a^2q^{12}\sigma^4 + 18816a^2q^{10}\sigma^6 + 117600a^2q^8\sigma^8 \\ & + 376320a^2q^6\sigma^{10} + 564480a^2q^4\sigma^{12} + 322560a^2q^2\sigma^{14} + 40320a^2\sigma^{16}\end{aligned}$$

Comparison

$$\text{mean gap} = -28aq^6\sigma^2 - 210aq^4\sigma^4 - 420aq^2\sigma^6 - 105a\sigma^8$$

$$\begin{aligned}\text{variance gap} = &-2688a^2q^{12}\sigma^4 - 88704a^2q^{10}\sigma^6 - 1165920a^2q^8\sigma^8 \\ &- 7008960a^2q^6\sigma^{10} - 18133920a^2q^4\sigma^{12} \\ &- 15805440a^2q^2\sigma^{14} - 1975680a^2\sigma^{16}\end{aligned}$$

$$\begin{aligned}\text{MSE gap} = &-3472a^2q^{12}\sigma^4 - 100464a^2q^{10}\sigma^6 - 1233540a^2q^8\sigma^8 - 7191240a^2q^6\sigma^{10} \\ &- 18354420a^2q^4\sigma^{12} - 15893640a^2q^2\sigma^{14} - 1986705a^2\sigma^{16}\end{aligned}$$

$$\begin{aligned}\text{Bias}(g_{\text{naive}})^2 = &\ 784a^2q^{12}\sigma^4 + 11760a^2q^{10}\sigma^6 + 67620a^2q^8\sigma^8 + 182280a^2q^6\sigma^{10} \\ &+ 220500a^2q^4\sigma^{12} + 88200a^2q^2\sigma^{14} + 11025a^2\sigma^{16}\end{aligned}$$